WORKED SOLUTION

STEP III 2004, Question 6

Pure • Recurrence Relations

STEP III 2004, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s3-q6

The key idea

The expression F_n is a quantity attached to each *consecutive pair* of terms, and the opening identity tells us exactly when it stays the same from one pair to the next. For the recurrence in (i) the second factor vanishes, so F_n is an invariant and we only need its value at the start. Part (ii) runs the argument backwards: if F_n is constant, the product must vanish, so at every step one of the two factors is zero. That gives a choice at each step, and the period-4 sequence comes from choosing well.

Solution

The identity

Writing out both terms, the w_{n-1}^2 terms cancel:

$$\begin{aligned} F_n - F_{n-1} &= (w_n^2 + w_{n-1}^2 - 4w_n w_{n-1}) - (w_{n-1}^2 + w_{n-2}^2 - 4w_{n-1}w_{n-2}) \\ &= (w_n^2 - w_{n-2}^2) - 4w_{n-1}(w_n - w_{n-2}) \\ &= (w_n - w_{n-2})(w_n + w_{n-2}) - 4w_{n-1}(w_n - w_{n-2}) \\ &= (w_n - w_{n-2})(w_n + w_{n-2} - 4w_{n-1}), \end{aligned}$$

which holds for $n \geq 2$ (we need w_{n-2} to exist), as required.

One more observation will be useful throughout: F_n is *symmetric* in w_n and w_{n-1}. Swapping them leaves $w_n^2 + w_{n-1}^2 - 4w_n w_{n-1}$ unchanged.

Part (i)

Take $w_n = u_n$, and let F_n be the corresponding quantity. For $n \geq 2$ the recurrence says $u_n + u_{n-2} - 4u_{n-1} = 0$, so the second factor in the identity is zero and

$$F_n = F_{n-1} \quad (n \geq 2).$$

Hence $F_n = F_1$ for every $n \geq 1$, and

$$F_1 = u_1^2 + u_0^2 - 4u_1u_0 = 4 + 1 - 8 = -3.$$

So $u_n^2 + u_{n-1}^2 - 4u_nu_{n-1} = -3$, that is, $u_n^2 + u_{n-1}^2 = 4u_nu_{n-1} - 3$ for all $n \geq 1$, as required. (The sequence runs $1, 2, 7, 26, 97, \ldots$.)

Part (ii)(a)

Putting $n = 1$ and $v_0 = 1$ into $(*)$:

$$v_1^2 + 1 = 4v_1 - 3 \iff v_1^2 - 4v_1 + 4 = 0 \iff (v_1 - 2)^2 = 0,$$

so $v_1 = 2$. There is no choice here, because it is a repeated root.

Now take $w_n = v_n$. Condition $(*)$ says precisely that $F_n = -3$ for every $n \geq 1$. So for $n \geq 2$ both F_n and F_{n-1} equal -3, and the identity gives

$$0 = F_n - F_{n-1} = (v_n - v_{n-2})(v_n + v_{n-2} - 4v_{n-1}).$$

A product of two real numbers is zero only if one of them is zero, so either $v_n = v_{n-2}$ or $v_n = 4v_{n-1} - v_{n-2}$, as required.

Ask yourself Is this all? Do both options really give a valid next term?

Yes. For fixed v_{n-1}, condition $(*)$ is the quadratic $x^2 - 4v_{n-1}x + (v_{n-1}^2 + 3) = 0$ in $x = v_n$. We know $x = v_{n-2}$ is a root, because F is symmetric and $F_{n-1} = -3$. The roots add up to $4v_{n-1}$, so the other root is $4v_{n-1} - v_{n-2}$. So the two options in (a) are exactly the two roots. We may pick either at each step, and the result always satisfies $(*)$. (When the roots coincide, as at $n = 1$, there is only one option.)

Part (ii)(b)

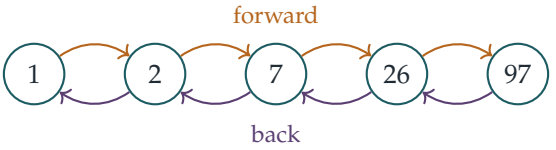
Here $v_0 = 1$, as required, and every consecutive pair (v_{n-1}, v_n) is either $(1, 2)$ or $(2, 1)$. Since F is symmetric, both give

$$v_n^2 + v_{n-1}^2 - 4v_nv_{n-1} = 1 + 4 - 8 = -3,$$

so $(*)$ holds for every $n \geq 1$. In the language of (a), this sequence always takes the option $v_n = v_{n-2}$.

Part (ii)(c)

We have $v_0 = 1$ and $v_1 = 2$. By (a), each later term is either the term two places back ("step back") or $4\times$(previous) $-$ (the one before) ("step forward"). From $(1, 2)$, stepping forward gives 7, then 26, 97, and so on, which is the sequence u_n. Stepping back undoes the last step forward.



So the sequence is a walk along this chain, starting at 1 and moving to a neighbour each time. Since 1 has only one neighbour, the walk must go to 2 straight after visiting 1.

To get period 4 with $v_0 = 1$, we need $v_4 = 1$. The period-2 sequence of (b) just shuttles between 1 and 2. To do something new, we go one step further:

$$v_0 = 1, \quad v_1 = 2, \quad v_2 = 4 \cdot 2 - 1 = 7, \quad v_3 = v_1 = 2, \quad v_4 = v_2 \text{ or } 4 \cdot 2 - 7 = 1.$$

Choosing $v_4 = 1$, and then repeating, gives

$$v_n = \begin{cases} 1 & n \equiv 0 \pmod 4, \\ 2 & n \text{ odd}, \\ 7 & n \equiv 2 \pmod 4, \end{cases} \qquad \text{i.e.} \quad 1, 2, 7, 2, 1, 2, 7, 2, \ldots$$

To check $(*)$ directly: every consecutive pair is $(1,2)$, $(2,7)$, $(7,2)$ or $(2,1)$. By symmetry we only need

$$1 + 4 - 8 = -3 \quad \text{and} \quad 4 + 49 - 56 = -3,$$

which both hold. So $(*)$ holds for all $n \geq 1$, and the sequence has period 4 (and not period 2, since $v_2 \neq v_0$).

In fact this is the only answer of exact period 4. After $1, 2$, the walk needs $v_4 = 1$, so it can only reach 7 at v_2 and must come straight back.

Checks

- Compute u_n up to $n = 40$ and confirm that $u_n^2 + u_{n-1}^2 - 4u_n u_{n-1} = -3$ every time. For example, $(26, 97)$: $676 + 9409 - 10088 = -3$.
- Test the opening identity on thousands of random integer triples.
- List every sequence allowed by (a) up to v_8, starting $1, 2$. The only ones with $v_{k+4} = v_k$ throughout are $1, 2, 1, 2, \ldots$ and $1, 2, 7, 2, \ldots$.

Common slips

- Concluding $F_n - F_{n-1} = 0$ in (ii) without saying why. It is because $(*)$ makes *every* F_n equal to -3.
- Forgetting that v_1 is forced. The quadratic has the repeated root 2, so no case split is needed.
- In (c), checking only one period's worth of pairs and forgetting the "wrap-around" pair $(v_3, v_4) = (2, 1)$.

Beyond STEP

The quantity F_n is an *invariant* of the recurrence: a conserved quantity, like energy in mechanics. The equation $x^2 + y^2 - 4xy = -3$ is a Markov-type (or Pell-type) Diophantine equation. Swapping one root of the quadratic for the other, $x \mapsto 4y - x$, is the classic "Vieta jumping" move, which generates all its integer solutions from the smallest one. Davenport's *The Higher Arithmetic* covers Pell's equation and its solutions by recurrences.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.